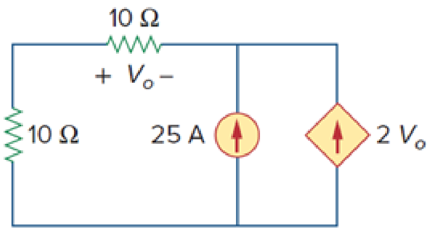
Chapter 2, Problem 22P

Problem

Find *Vo* in the circuit in Fig. 2.86 and the power absorbed by the dependent source.

Figure 2.86:



Step-by-step solution

Step 1 of 5

Refer to Figure 2.86 in the textbook.

Draw the circuit diagram with voltage notations.

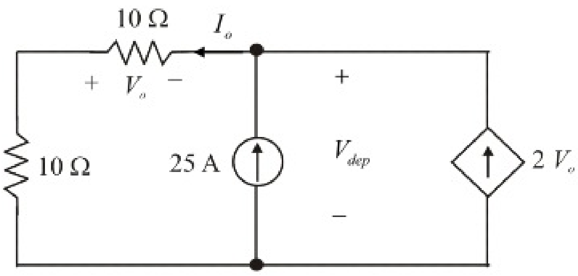


Figure 1

Step 2 of 5

Use Ohm's law to Figure 1, the voltage across top $10\ \Omega$ resistor is,

$V_o = -10I_o$ …… (1)

Apply Kirchhoff's current law to Figure 1 and calculate the current, i_o.

$$25 + 2V_o = I_o$$

Substitute $-10I_o$ for V_o.

$$25 + 2(-10I_o) = I_o$$
$$25 - 20I_o = I_o$$
$$21I_o = 25$$
$$I_o = \frac{25}{21}$$
$$I_o = 1.19\ \text{A}$$

Step 3 of 5

Substitute $1.19\,\text{A}$ for I_o in equation (1).

$$\begin{aligned} V_o &= -10(1.19) \\ &= -11.9\,\text{V} \end{aligned}$$

Therefore, the voltage, V_o is $\boxed{-11.9\,\text{V}}$.

Step 4 of 5

Apply Kirchhoff's voltage law to Figure 1.

$$-V_{dep} + 10I_o + 10I_o = 0$$
$$V_{dep} = 20I_o$$

Substitute $1.19\,\text{A}$ for I_o.

$$\begin{aligned} V_{dep} &= 20(1.19) \\ &= 23.8\,\text{V} \end{aligned}$$

Step 5 of 5

The power absorbed by the dependent source is,

$$P_{dep} = -V_{dep}(2V_o)$$

Substitute $23.8\,\text{V}$ for V_{dep} and $-11.9\,\text{V}$ for V_o.

$$\begin{aligned} P_{dep} &= -(23.8)((2)(-11.9)) \\ &= 566.44\,\text{W} \end{aligned}$$

Therefore, the power absorbed by the dependent source is $\boxed{566.44\,\text{W}}$.